Aiden Harrington

aidenharrington.com aidenkharrington@gmail.com linkedin.com/in/aidenharrington/

Technical Skills

Java Spring Python PostgreSQL React Node.js AWS Docker Terraform

Experience

- **Arbutus West** Remote, Canada
  Software Engineer (Part-Time) *Nov 2025 - Present*
  - Designed and implemented a real-time engine load management system in Python using a PID control algorithm, improving industrial generator efficiency under high-speed operating conditions.
  - Architected full software lifecycle for a Raspberry Pi-based control system, from requirements to simulation, validation, and hardware integration.
  - Owned all software design decisions and delivered weekly progress reports to stakeholders.

- **Benevity (Layoff, Company-Wide Restructuring)** Remote, Canada
  Software Developer 2 *Jul 2026 - Sep 2026*
  - Co-built the US phase of an automated nonprofit bank verification system in Java Spring Boot using AI-driven development with Cursor, spanning new and existing microservices and databases, replacing a manual process and designed to remove manual review from the majority of cases.
  - Engineered an idempotent verification flow with internal automated retries driven by downstream responses from a third-party bank authentication vendor, ensuring a zero-PII architecture through production readiness.
  - Owned end-to-end delivery of enhancements to banking CSV generation and atomic transaction reversals across distributed databases, configuring cloud infrastructure via Terraform and AWS.

- **PayPal** Remote, Canada
  Software Engineer 2 *May 2022 - Mar 2024*
  - Contributed to the continuous delivery of Java Spring Boot features that reduced partner onboarding by 75% and merchant onboarding by 50%.
  - Led technical delivery of a cross-functional webhook system to alert merchants at threshold criteria, aligning product, architecture, and partner stakeholders.
  - Secured partner approval and led release demos, enabling real-time merchant notifications.
  - Scaled Spring Batch tool from proof-of-concept to production, redesigning process flows and implementing automated retry mechanism, enabling successful rollout for 100k+ accounts.
  - Led contract engineers on multiple projects: defined requirements, ensured timely delivery, and maintained code quality via PR reviews.
  - Mentored summer intern: designed project, provided daily guidance and supported successful demo completion.

- **Knoldus** Remote, Canada
  Software Intern *May 2021 - Aug 2021*
  - Developed full-stack features for an internal employee leaderboard app spanning Angular frontend, Scala REST endpoints, and PostgreSQL schema/query design, with unit test coverage throughout.
  - Led two other interns by assigning Sprint tasks, performing code reviews, providing technical guidance, and presenting weekly updates to the CEO and Manager.

- **Mitacs** Vancouver, BC
  Junior Software Developer (Co-op) *Sep 2019 - Aug 2020*
  - Built full-stack features for Mitacs' grant application portal — a complex form-driven system where project type dynamically determined team composition, funding eligibility, and role restrictions — using Angular and Node.js across frontend, backend, and database layers.
  - Developed a scheduled daemon to automate weekly data synchronization across the Research Portal, SharePoint, Azure, and MySQL, ensuring data consistency across four integrated systems.

Education

- **University of British Columbia** Vancouver, BC
  BSc, Computer Science *May 2022*

Projects

- **AiDB** *askaidb.com*
  - Launched a live web platform that transforms Excel files into online databases, enabling users to query data directly in SQL or through plain-English prompts translated by generative AI.
  - Architected a multi-cloud deployment across Firebase (hosting/auth), Firestore, AWS EC2, and RDS, with OpenAI API integration handling natural language → SQL translation at query time.
- **Subscribr** *https://www.youtube.com/watch?v=1xO1CIoY_48*
  - Created a full-stack subscription application packaged in a self-contained Docker container orchestrating a React frontend, two Java Spring Boot services, and a PostgreSQL database.
  - Implemented hybrid communication with synchronous API calls and event-driven updates (webhooks and emitters) to ensure reliable real-time data synchronization.